MATH 146: Calculus II
Winter 2026

Instructor Information

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Office Hours: After Class or by Appointment

Class Information

Dates: January 5 – March 15
Time: Period 3: 10:40 – 11:50 a.m.
Classroom: Science + Math Center - TBD

Course Description

Topics include a brief review of differential calculus and the fundamental theorem, applications of and techniques for evaluating the integral, including elementary multivariate integrals. Also includes Taylor series and further discussion of differential equations.

Course Objectives

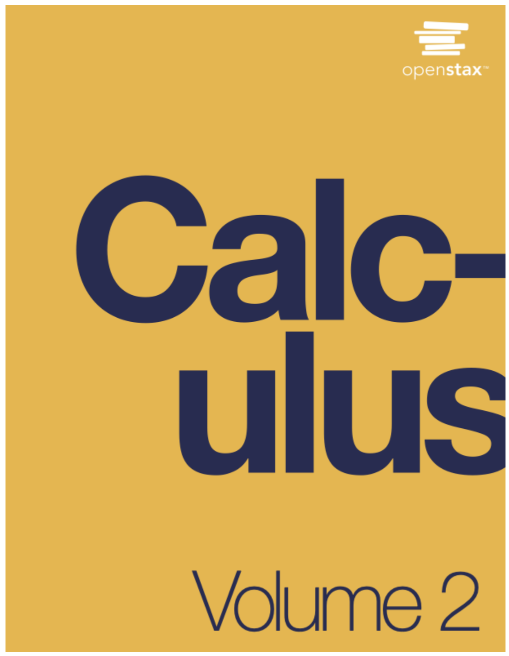
Students who complete this course will be able to:

1. Compute definite integrals using a wide range of techniques.
2. Apply definite integrals to solve common applications.
3. Determine when infinite sequences and series are convergent, and be able to evaluate them when possible.
4. Work with Taylor and power series, using them to approximate functions and solve calculus-related functions.
5. Describe curves parametrically and with polar coordinates and solve calculus related problems on these curves.
6. Solve basic differential equations and associated physical problems with them.
7. Use technology and numerical techniques to find approximate solutions to integrals and initial value problems.

Textbook, Calculators, & Software

Textbook: The textbook for the course will be a free online textbook "Calculus 2" (https://openstax.org/details/books/calculus-volume-2/) from OpenStax. (You can also print off a PDF file.) We will be covering most of the sections in the book, though not necessarily in the order covered in the textbook.

This textbook is part of a series from the same publisher. If you need to review material from Calculus I, you can take a look at the Calculus 1 textbook (https://openstax.org/details/books/calculus-volume-1).



Calculator: You may need a calculator to do the computations that will arise throughout the course. No specific calculator is required; I recommend the TI-30XS MultiView (https://a.co/d/0QHmuee). It's cheap and should do anything you need to in this course. You don't need a graphing calculator for this course, but if you have a TI-84 already, it's more than capable.

Class Attendance and Participation

It is essential to your success in this course that you attend and participate in each lecture. I will not take attendance directly, however there is an element of your grade that will include in-class elements. In my experience as a student, a teaching assistant and as a lecturer, the number one most common problem students have in courses is falling behind in the course and not being able to catch back up. If you are forced to miss a day, it is your responsibility to catch yourself back up, whether through getting notes from a classmate or coming to me to go over the material.

In class, I ask that you respect my time, your time and your classmate's time. In class, please avoid anything that would be too distracting your classmate: no cell phones, music, loud foods, etc. I personally recommend taking notes with pen and paper, but if you prefer using a laptop/tablet, please stay focused on the lecture and don't go surfing the web.

I aim to create an open and welcoming environment for learning, so if you have questions or don't understand something I've said, please speak up. If you are confused about something, it's almost certain that you're not the only one, and I'm happy to clarify. If you notice a mistake in my math, let me know, I'm only human and I promise I've made more stupid math mistakes than you have! And if you're curious or want to know more about a subject, speak up, I got my PhD in Mathematics & Statistics because I love this stuff!

Email Policy

If you have any questions outside of class, you can stop by my office, but you can also email me. I will attempt to respond to all emails within 24 hours. If you send an email late at night, I may not respond until the next day, and if you email on the weekends, I may not respond until the following Monday. If I fail to respond after 24 hours during the week, please resend the email, as I may have just not noticed it. I recommend that you email me using your Knox College email account, and keep note that there may be topics I may not be able to discuss over email or feel comfortable discussing over email, in which case I will make a time for us to meet in person to speak.

Academic Integrity & Honor Code

Remember that all work you submit must be in accordance with the Knox College Honor Code (https://www.knox.edu/offices/academic-affairs/honor-code-and-procedures/). Should I find evidence of a violation of academic integrity, I will sanction you.

Note two passages from the Honor Code in particular:

> *Students are expected to leave books and notes elsewhere during an examination and to take examinations in public places such as classrooms or open lounge areas nearby in the buildings. Washrooms, storage areas, and maintenance areas in the basement and elsewhere are not public places. Neither are carrels or other closed areas . . .*
>
> *Faculty may require that students not bring books, notes or electronic devices into the building in which an examination is administered. Alternatively, they may permit students to bring these resources into the building but may require that these books, notes and electronic devices be left in a central area or the classroom throughout the duration of the examination or other academic obligation.*

In accordance with these policies, you should not have a cell phone on you when taking exams or quizzes. I will not allow computers or other electronic devices, notes, textbooks or any other help on exams.

Homework can be done cooperatively, but any submissions must be in your own words.

AIs such as Chat-GPT are expressly forbidden in solving the problem, both because they will not help you learn (the only way to learn mathematics is through doing mathematics), and because it doesn't even answer correctly most of the time, so don't waste your time. I am open to you using is as a learning aid, and would even be interested in hearing *how* you all use it in your studying, but you need to be able to solve the problems on the quizzes and the exams, so do not rely on it!

Office of Disability Support Services

The Office of Disability Support Services website can be found at https://www.knox.edu/offices/disability-support-services.

Knox College abides by Section 504 of the Rehabilitation Act of 1973 (https://www.epa.gov/external-civil-rights/section-504-rehabilitation-act-1973), which stipulates:

> *No otherwise qualified individual with a disability in the United States, as defined in section 7(20) shall, solely by reason of her or his disability, be excluded from the participation in, be denied the benefits of, or be subjected to discrimination under any program or activity receiving Federal financial assistance or under any program or activity conducted by any Executive agency or by the United States Postal Service.*

Disabilities covered by law include, but are not limited to, learning disabilities, psychological disabilities, health impairments, hearing, and sight or mobility impairments.

If you have a disability that may have some impact on your work in class and for which you may require accommodations, please contact Stephanie Grimes in the Office of Disability Support Services (office: SMC E-115; email: sgrimes@knox.edu) so that such accommodations may be arranged.

Note that the accommodations are not retroactive; I do not make adjustments until ODSS contacts me to let me know what I can do. Also note that accommodations must be renewed each term.

Tutoring

The Center for Teaching and Learning (CTL) offers many resources for students in need of help. Asking for help is no reason to be embarrassed; seeking help in a big part of the learning process, and a sign that you are engaging with the material and your coursework. You are not alone at Knox College and you should utilize all the tools at your disposal to succeed.

Writing Tutors: If you want help on a paper, no matter the course, you can find writing tutors in the Writer's Workshop, which is at the Center for Teaching and Learning (466 S. West St., across from Post Hall), Monday through Friday, from 10:30 to 4:00pm, and again in Red Room on Tuesday, Wednesday, and Thursday nights from 7:00 to 9:00pm. You can also schedule appointments to meet with a writing tutor either online or in person (should you both so desire) by visiting the Center for Teaching and Learning website.

Subject Tutors: The Red Room study tables are great resources for students wanting help with virtually any subject— including MATH 145. There are two Red Room study areas: "Red Room" and "Red Room SMC." Red Room SMC is intended to support most classes taught in SMC and is located in the Learning Commons in SMC (under the whale), while Red Room is for most other courses and is located in the Red Room on the second floor of Seymour Library. Both study areas are free and open on a walk-in basis from 7:00pm to 9:00pm, on Tuesday, Wednesday, and Thursday nights. Specifically, this class, MATH 145, is tutored in the Red Room in SMC.

TimelyCare

If you're unfamiliar with mental health support, taking care of your mental well-being can be stressful, or even scary. Knox is partnered with TimelyCare to provide a virtual health care solution that reflects Knox's values and student population. It is a student-centered platform that offers on-demand access to mental healthcare and a diverse, culturally competent provider network. TimelyCare complements Knox's on-campus services with 24/7 on-demand virtual mental health care, health coaching, peer to peer support and a plethora of coping strategies ranging from homesickness to depression to study skills. There are 12 free sessions available from September to August annually and can be accessed anywhere on or off campus (even internationally!). To access TimelyCare, go to TimelyCare.com or use the QR code below, register with your Knox email and follow prompts! If you have any questions, please reach out to Knox Health and Wellness at counseling@knox.edu or (309) 341-7492. If there is an emergency, please call 911 or contact Campus Safety at (309) 341-7979.

Title IX

Knox College values a culture of respect. If you have encountered any form of discrimination or harassment, you are encouraged to report it at https://www.knox.edu/about-knox/our-values/culture-of-respect/report-it.

As a faculty member, I am required to disclose to the Director of Title IX & Civil Rights Compliance any information I receive about discrimination, harassment, sexual misconduct and other forms of interpersonal violence. This includes information shared in class assignments, discussions, emails, and/or shared conversations outside of class. If a disclosure is received, she will reach out to you. You are not required to respond to this outreach.

If you wish to speak with a confidential resource (one not required to report to Title IX), please contact:

- Knox College Health Services (https://www.knox.edu/offices/health-and-counseling-services/health-services)
- Knox College Counseling Services (https://www.knox.edu/offices/health-and-counseling-services/counseling-services)
- Knox College Director of Spiritual Life (https://www.knox.edu/offices/spiritual-life/meet-the-staff)
- TimelyCare (https://app.timelycare.com/auth/login)

Pregnancy and/or Pregnancy Related Conditions

Faculty members are required to inform a student who is pregnant or experiencing a pregnancy related condition (including termination) that the Director of Title IX & Civil Rights Compliance is available to provide support by helping you to understand your rights/options and provide supportive measures. You can contact the Director of Title IX & Civil Rights Compliance at cultureofrespect@knox.edu.

Grading Information

Exams

We will have two midterm exams in this course, tentatively in the 4th and the 8th week of the course. The exact date will be confirmed as we approach the date. Each test will account for about **15% of your grade**, for a total of **30% of your grade**.

At the end of the course, we will have a cumulative final exam during the time scheduled by the registrar's office. It will count for about **20% of your grade**.

Exams will be given in the classroom that we do lectures in. I have spoken with Dean Crawford, and I will be expecting everyone to take the exam in the room, unless you have an accomodation and prior approval. A calculator is all you can use on the exam; no computers, no phones, no notes, not talking to other classmates.

Homework Assignments

Homework will be done as a combination of WebWork assignments (WILLGETTHISWHENIT'SMADE), pencil-and-paper assignments and possibly programming assignments in Python. I will assign them throughout the course. The assignments will total to about **25% of your grade**. While you do the WebWork assignments, write up your work neatly and clearly in a "problem journal" you can use to study from. I may ask to check this work for credit.

Quizzes

On Thursdays that are not exam days, I will set aside some time for any questions you have on the weeks material, and then I will give a quiz. These quizzes will count for a total of around **15% of your grade**.

In-Class Activities

During most lectures, I will try to assign small assignments, likely a practice problem done together or a small "quiz". These will total to **10% of your grade** overall. I will drop enough of these that you will have a few days to be able to miss, but I urge you in the strongest possible way to please *please* attend all lectures.

Grading Breakdown

Thus the grades will break down approximately as follows:

Component	Points
Midterm Exam #1	15%
Midterm Exam #2	15%
Final Exam	20%
Homework	25%
Quizzes	15%
In-Class	10%

Grade Scale

Final grades will be assigned according to the following percentages, though I reserve the right to alter the ranges as need be:

Grade	Range
A	>93%
A-	90% – 93%
B+	87% – 90%
B	83% – 87%
B-	80% – 83%
C+	77% – 80%
C	73% – 77%
C-	70% – 73%
D+	67% – 70%
D	63% – 67%
D-	60% – 63%
F	<60%